Preliminary report

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1. Abstract

The objective of the project is to develop a miniature wearable device that can power a thread-like stimulating implant designed by the researchers at the Neural Interface Technology Laboratory, led by Dr. Adam Khalifa. The system architecture of the power transmitter device implements the first-of-its-kind battery-powered radio frequency (RF) embedded system on a custom PCB that generates a 25 MHz differential pulsed signal at 1 W delivered to the implant through needle electrodes. The standby current of the device is only 1 mA, enabling long battery life. In addition to enabling neural stimulation, the device also records real-time single-channel EMG at a sampling rate of 1 kHz to verify if the stimulation produced the desired effect. The RF power transmitter functionality was verified using a spectrum analyzer, while EMG recording was verified using an artificial signal supplied by a signal generator into a saline medium mimicking muscle. The device was first prototyped using evaluation kits, then integrated on a modular, easily-debuggable board. After refining the design using the learnings from these versions of the system, a miniature and highly integrated implementation was designed and fabricated on a 65 mm x 48 mm PCB. Upon the conclusion of the testing, the prototype will be delivered to a partner laboratory at Harvard University.

2. Introduction

Implantable Medical Devices (IMDs) need to be powered externally after being surgically implanted in the patient's body. Differential tissue-coupled powering is a novel method of powering IMDs developed at the Neural Interface Technology Lab. It provides an alternative to already existing techniques such as TENS [1], with improvements in the areas of efficiency, durability, and invasiveness compared to state-of-the-art [2-10]. The transmitter-recorder device coupled with the stimulating implant completes a full medical system:

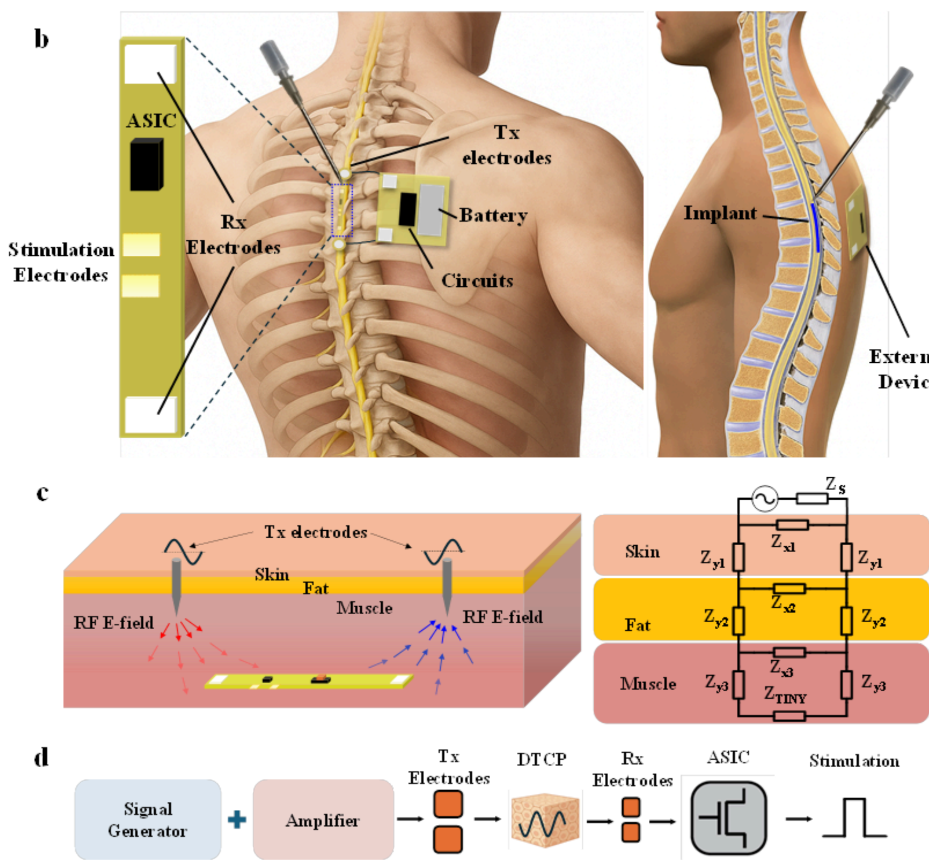


Figure 1: Overview of the DTCP system. In our work, we are focusing on developing the external powering device. Source: DTCP journal paper draft, Han Wu, University of Florida, Neural Interface Technology Lab [15].

The external transmitter device needs to be a miniature embedded system capable of pulsed powering and with low standby current, enabling long battery life. It will be designed leveraging already existing “building block” integrated circuits that can be connected together to form a full embedded RF solution. Based on the experimental results achieved with benchtop setups, Dr. Adam Khalifa suggested the following specifications to be fulfilled by the device:

Performance metric	Target value
Output frequency	25 MHz
Output power	≥30 dBm differential
Static current	<10 mA
Tx current	<1000 mA
EMG noise floor	<0.5 mV
EMG sampling rate	≥1000 samples/s
Board dimensions	50 mm x 70 mm

Figure 2: Table of target design specifications for the project.

3. Background

A 25 MHz signal cannot be generated by a simple application of a built-in DAC on a microcontroller, as that would require at least 50 MHz system clock speed, which would significantly increase the cost and power consumption of the microcontroller solution [11]. Hence, the nature of the application demands an external signal generator that can be shut down by the microcontroller to lower the average current consumption. The AD9850 direct digital synthesizer (DDS) with an external 100 MHz crystal was chosen to fulfill that requirement. The maximum output power of that device is 8 dBm. Hence, the power amplifier used for the DDS signal needs to provide at least 22 dB of gain and saturate at at least 30 dBm of power to achieve the project requirements. However, all the previously published designs of power amplifiers for this frequency range usually target way higher output power than what’s required by this project, which leads to a large form factor and low gain and efficiency [12,13]. A search of off-the-shelf power amplifiers on Digikey and Mouser revealed that a two-stage broadband design is necessary to meet the project requirements.

EMG is a type of electrical signal generated by muscle activity, like muscle contraction. The frequency of this type of signal is up to 500 Hz, and the amplitude ranges up to 10 mV [14]. It is measured using electrodes, which in our case are needles inserted into the measured muscle. In the case of the DTCP external powering device, EMG is used to obtain nerve stimulation feedback - after the implant delivers a stimulation signal to the muscle’s nerve, researchers need information about whether the stimulation was successful.

4. Design

The high-level system design of the transmitter-recorder device is shown in the figure below:

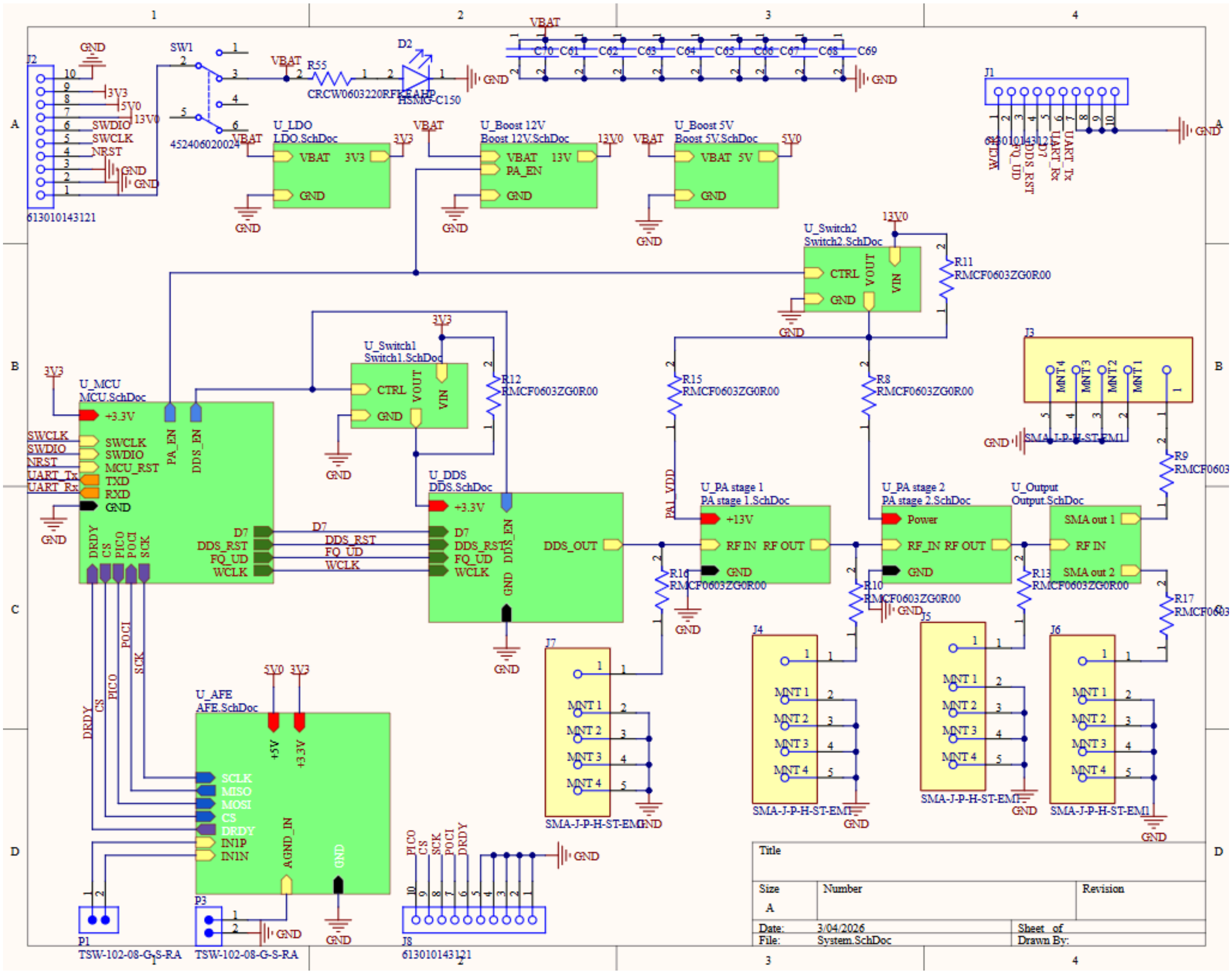


Figure 3: system schematic of the final version of the device.

The 2D and 3D views of the PCB layout are shown below:

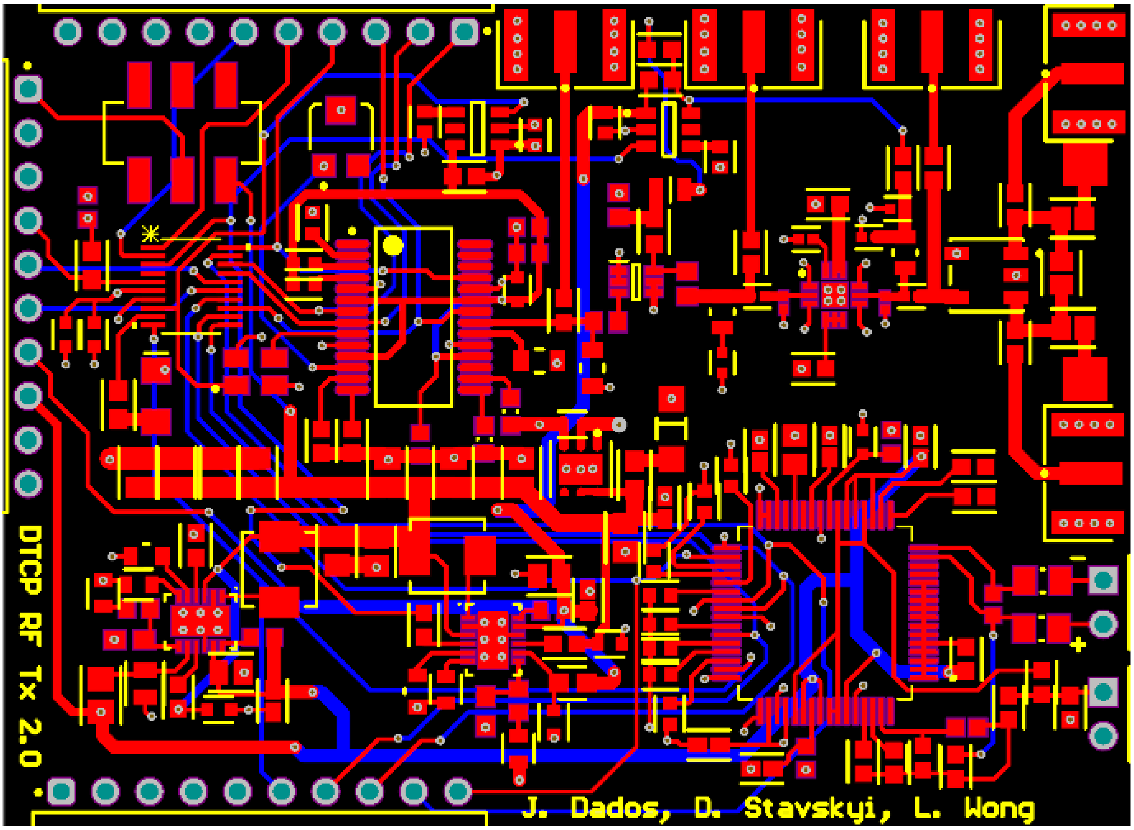


Figure 4: 2D PCB layout.

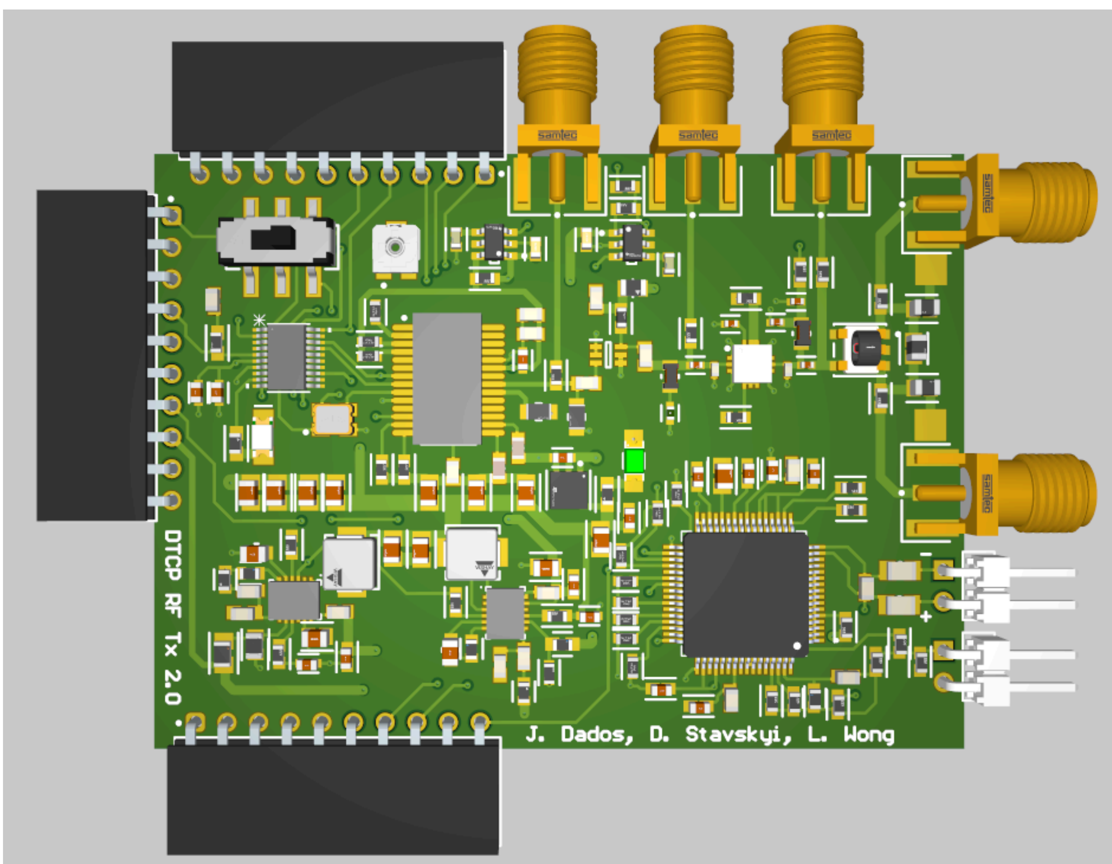


Figure 5: 3D render of the assembled board.

In the following subsections, each sub-system of the device will be described in detail.

a. RF Signal Generation
Review of electronic RF signal generator ICs available on the market revealed that a DDS would be the best solution based on how easy it is to digitally tune its output frequency. Other options were rejected for the following reasons:

- VCOs: lower output power (below 0 dBm), harder frequency control, fewer options available in the HF band.
- Crystal oscillators: no frequency control, not designed for RF signal generation.

After a careful search of available DDS options, AD9850 was chosen due to its ability to deliver signals up to 40 MHz.

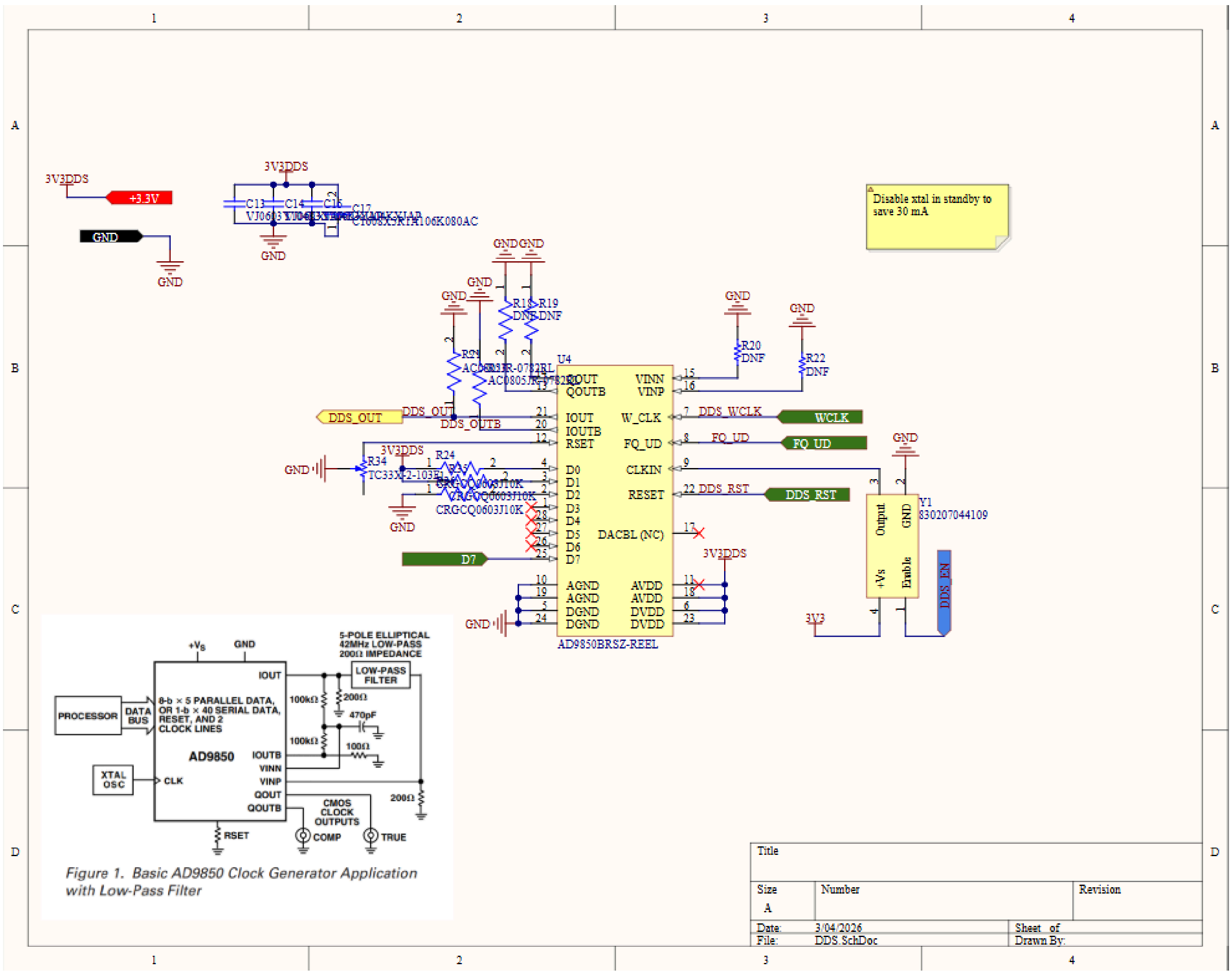


Figure 6: DDS sub-system schematic.

However, the power consumption of the DDS being around 60 mA would significantly shorten the battery life if the DDS were powered on all the time. Because of that, a power switch has to be used by the microcontroller to turn off the DDS during the periods of time when power transmission is not needed.

b. Power switching

The part choice criteria for the power switch were as follows:

- 0 ~ 12 V input and output voltages.
- >2A output current.
- Simple typical application circuit.

After a Digikey and Mouser search, TPS22810-Q1 was chosen as the most promising candidate based on its datasheet specs combined with the simplicity of use. Each power switch only requires two external components:

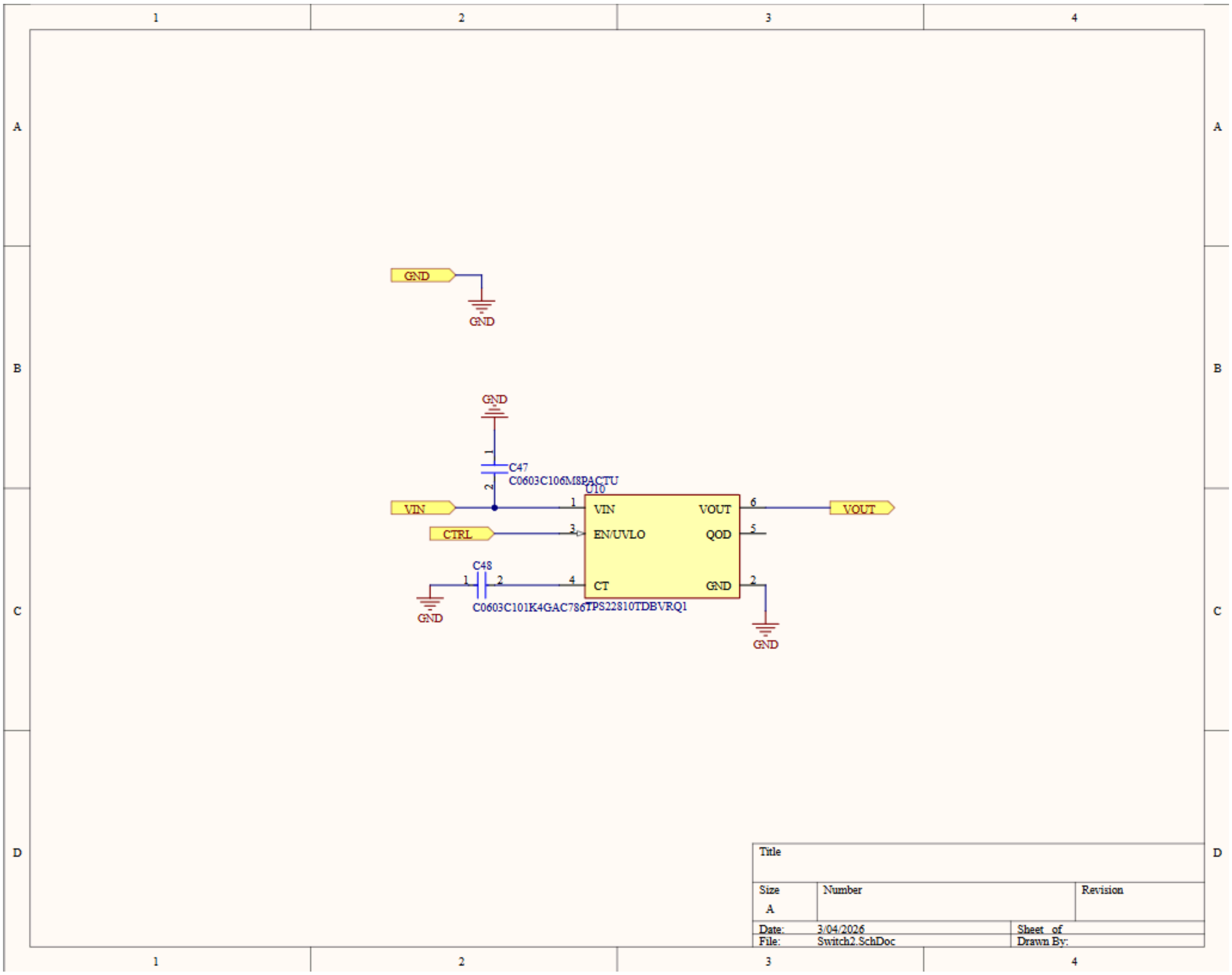


Figure 7: DDS sub-system schematic.

c. Power amplification

We assumed that the real output power of the DDS on the PCB will likely be less than the theoretical value. Because of that, the PA was designed to work with input signals as low as 5 dBm. Different topologies, such as Class-A and Class-E, were considered in the design process. After extensive testing of a custom class-E solution, we decided to use a two-stage solution: TQP369184 by Qorvo as the first stage, followed by PMA3-43-1W+ by Mini-Circuits as the second stage.

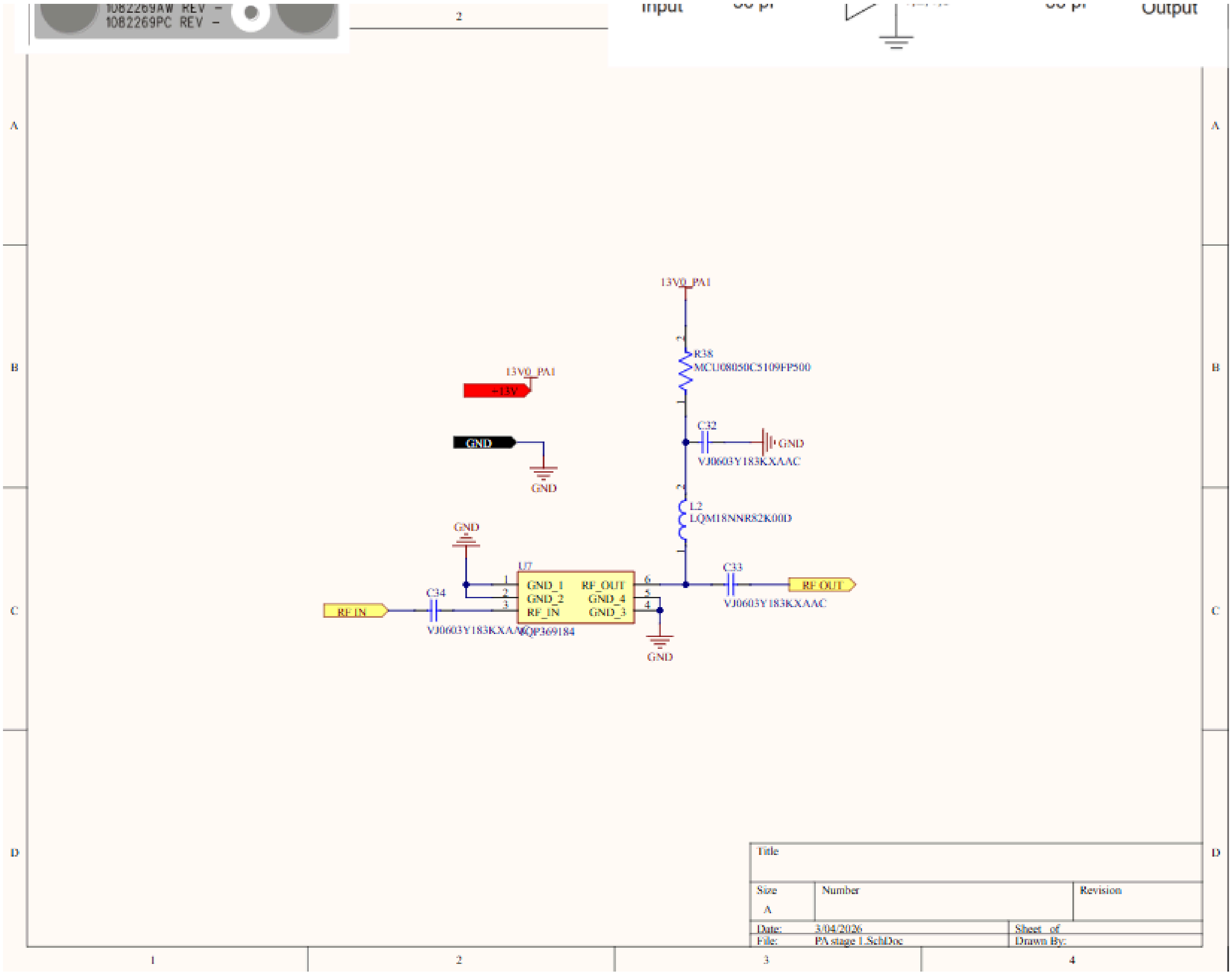


Figure 8: First-stage PA sub-system schematic.

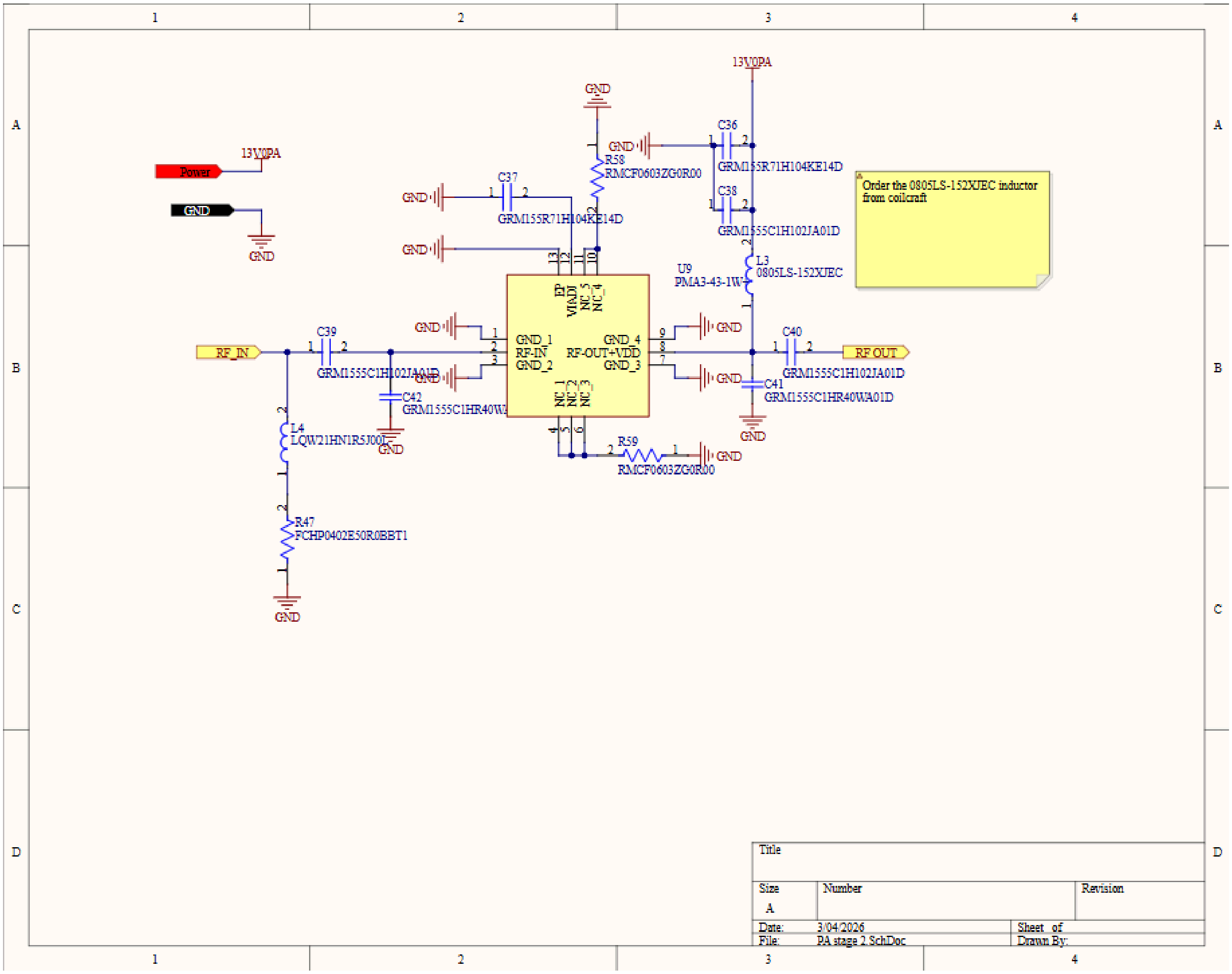


Figure 9: Second-stage PA sub-system schematic.

d. EMG recording

The choice of the AFE (analog front-end) chip was a critically important part of the design. The following criteria were used:

- programming interface
- part cost
- sampling rate
- amplifier bandwidth

In the end, ADS1299 by Texas Instruments was chosen due to offering the best balance of the above characteristics.

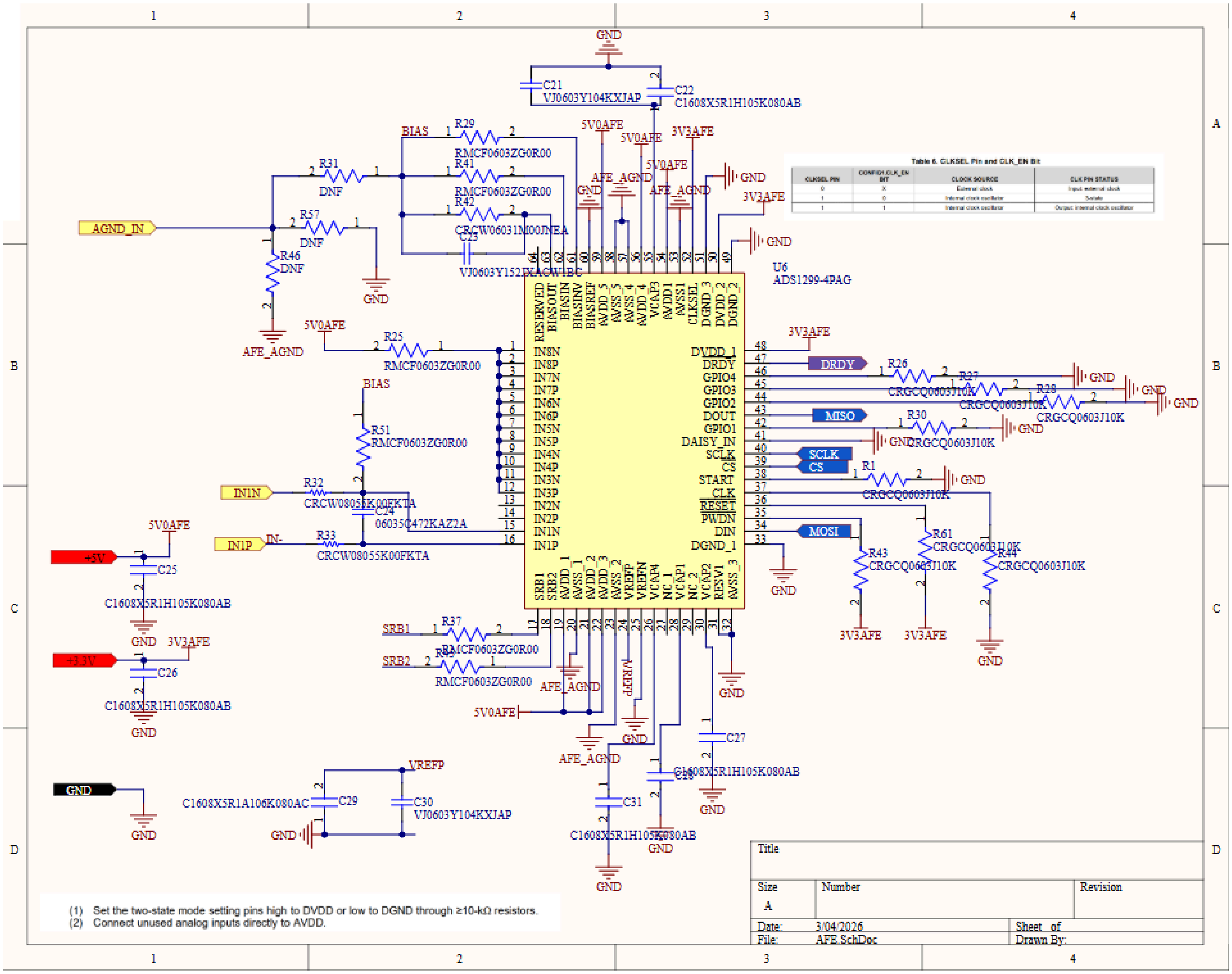


Figure 10: AFE sub-system schematic.

e. Microcontroller

When trying to find a suitable microcontroller for this embedded system, the main criteria of the search were the following:

- Minimal external circuitry needed for programming and operation.
- Minimal part cost.
- Minimal current consumption.
- Sufficient number of GPIO pins to support DDS programming, communication with the AFE, and control of power switches.

The search was conducted among Texas Instruments ARM-based microcontrollers. That family of devices was chosen due to our familiarity with TI firmware development environments and evaluation platforms. Ultimately, the MSPM0C1104 device was chosen due to its small footprint and minimal external circuits required. Furthermore, only 3 pins (SWCLK, SWDIO, NRST) with no external ICs on board would be needed to program the device using a LaunchPad. The microcontroller datasheet was used to verify that it can be powered directly from the battery or through a 3.3V LDO (preferred), and the expected current consumption is below 1 mA.

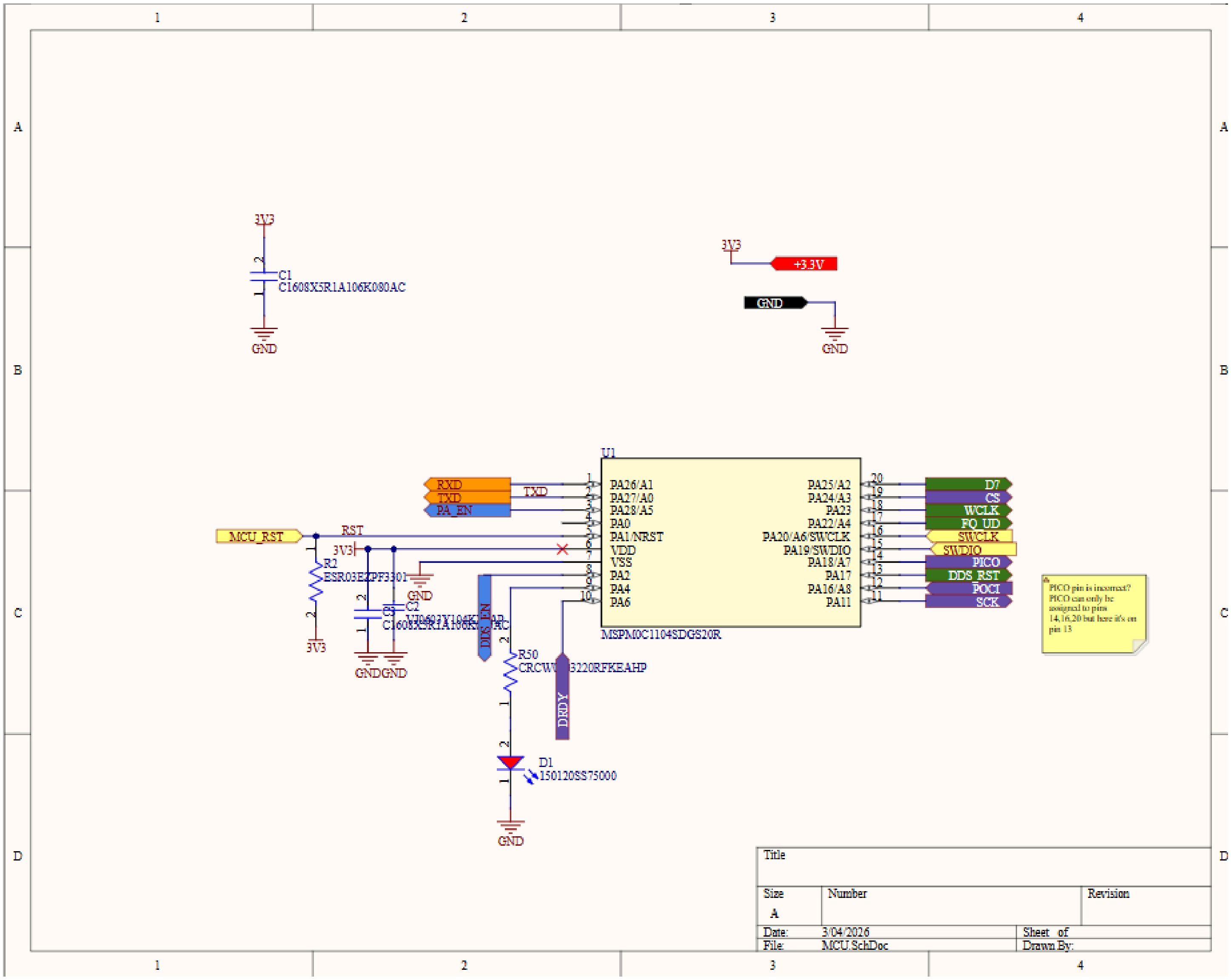


Figure 11: MCU sub-system schematic.

f. Power supplies

The requirements for the power circuit of the device were as follows:

- At least 2A output current limit for all power ICs
- LDO needed to drop 3.7V to 3.3V for the MCU and EMG AFE
- Boost converter needed to obtain 5.0 V and 13.0 V from the battery for the EMG AFE and the power amplifier.

Two instances of the LTC3122 boost converter were used to generate a 13.0 V and 5.0 V DC outputs to deliver power to the amplifiers and the AFE.

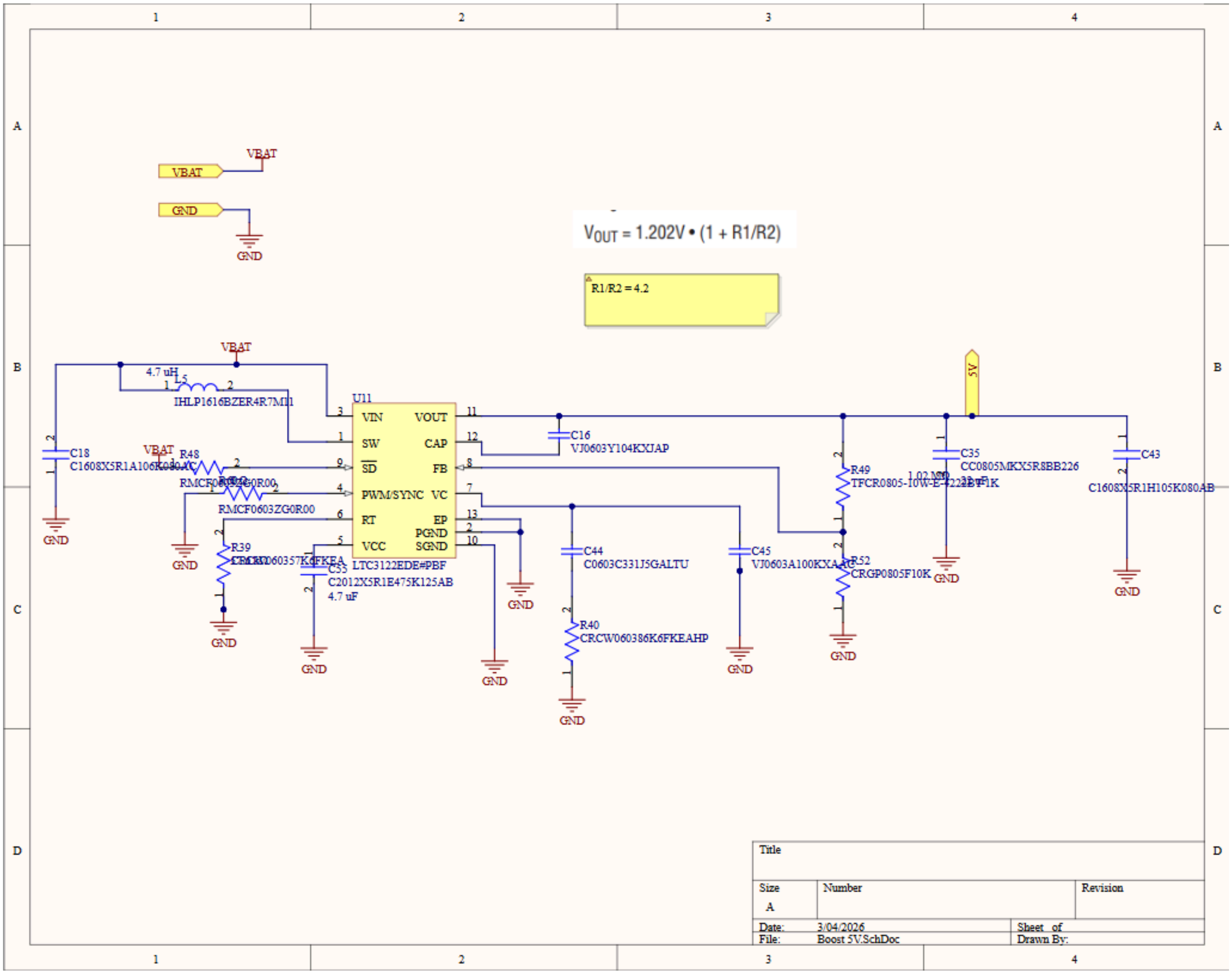


Figure 12: The 5V boost converter schematic.

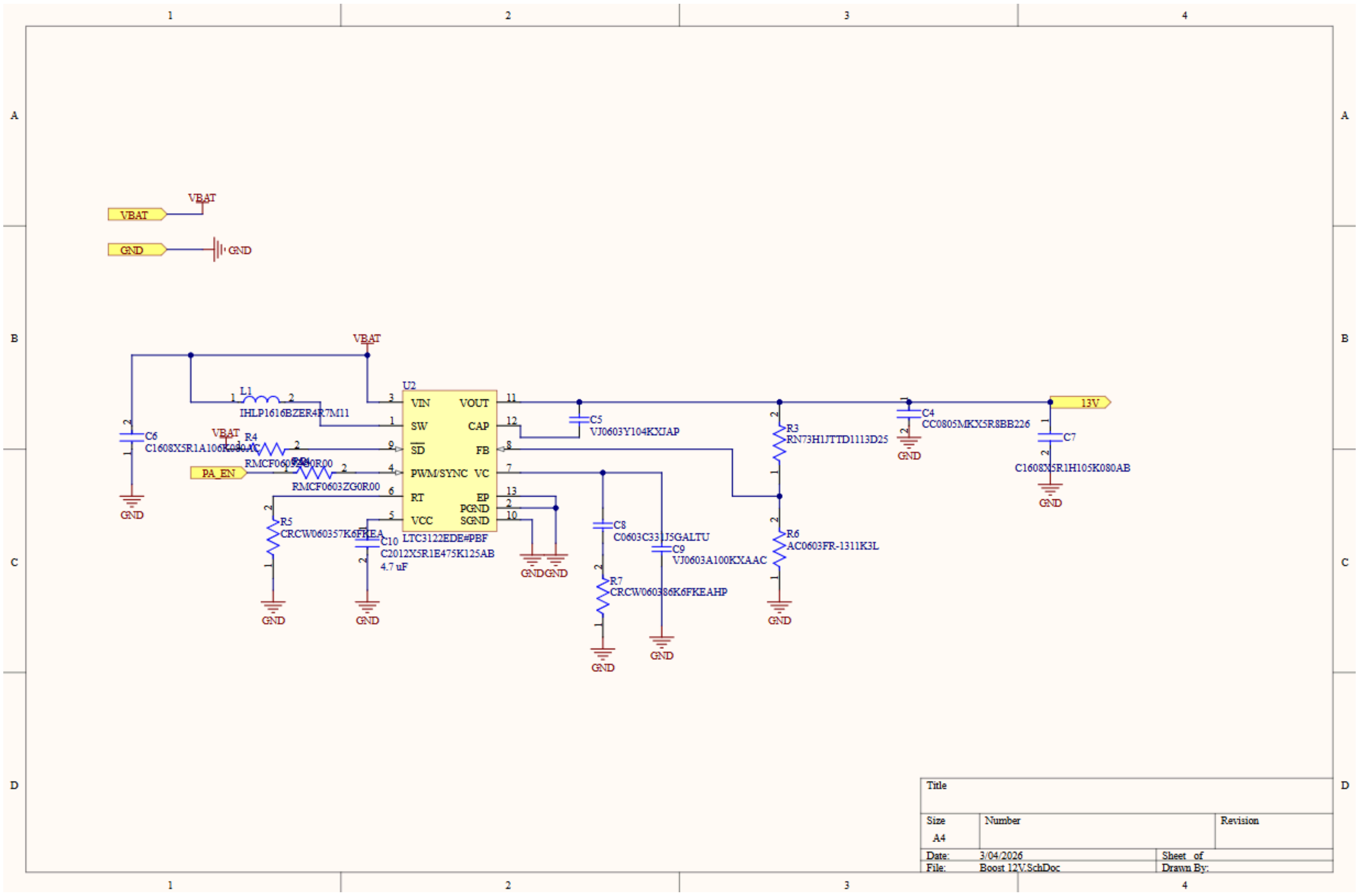


Figure 13: The 13V boost converter schematic.

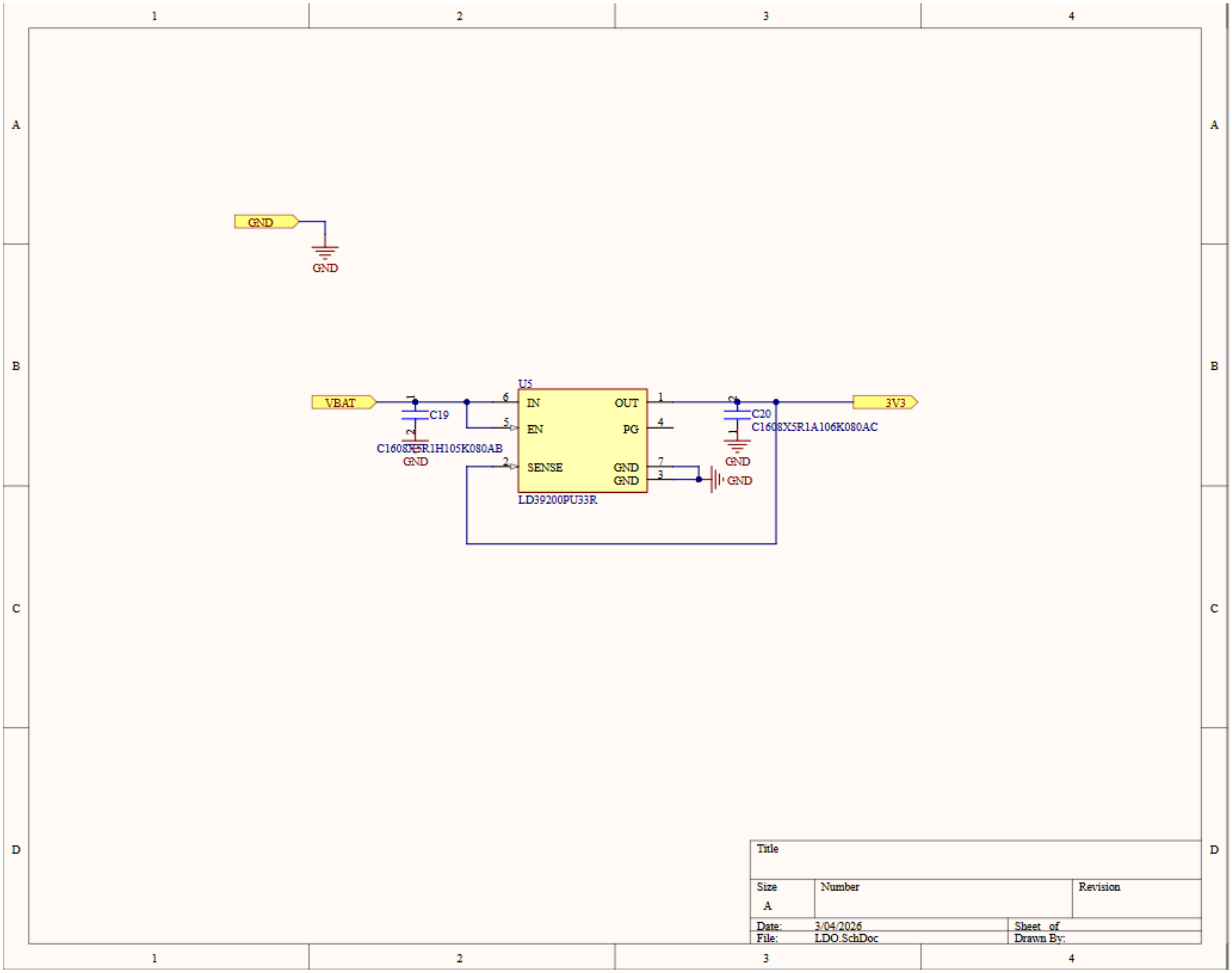


Figure 14: The 3.3V LDO schematic.

g. Communication between internal systems
The CC1310 microcontroller uses SPI to communicate with the AFE and a custom digital interface outlined in the DDS' datasheet to communicate with AD9850. It also used GPIO digital signals to control the power switches on the board.

h. External interfaces
The CC1310 Lauchpad mediates the connection between the CC1310 microcontroller and the user PC, allowing for flashing and debugging the firmware in real time as well as receiving the EMG readings over UART to be captured by the SerialPlot application.

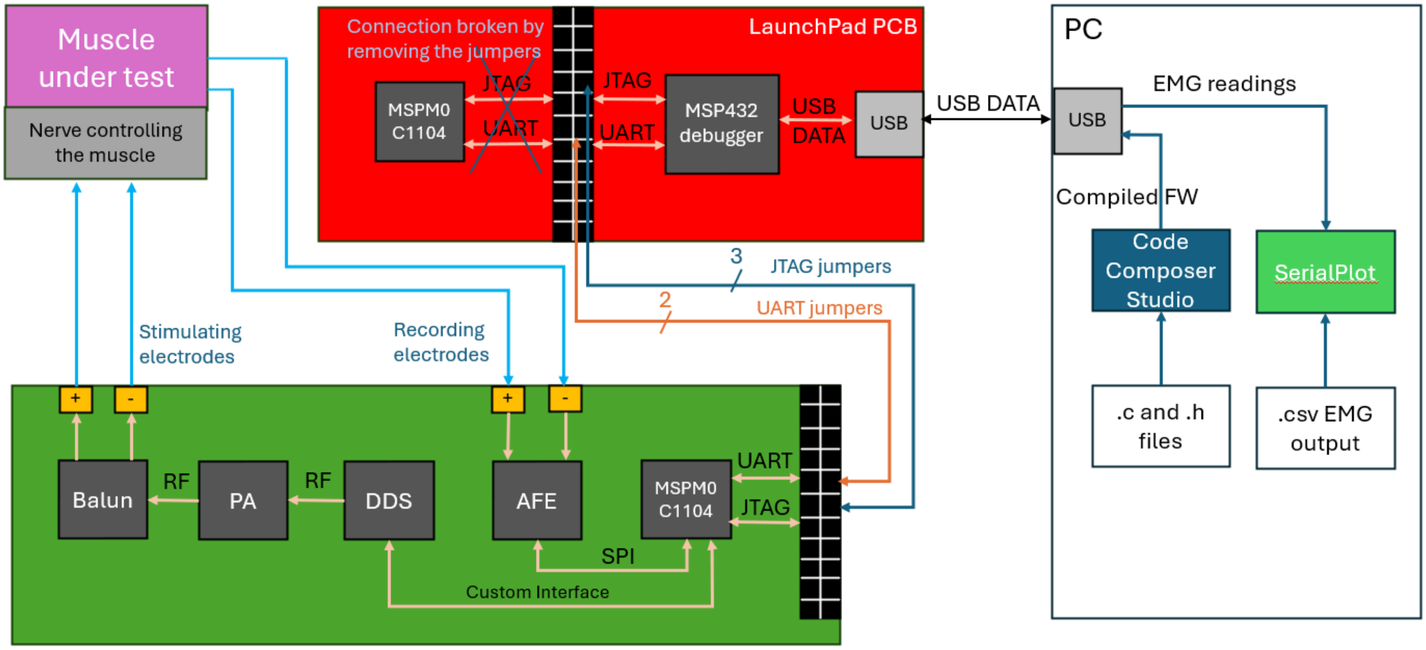


Figure 15: Overview of all internal and external interfaces of the device.

5. Tools, Design Constraints, and Engineering Standards

a. Tools

The PCB schematic layout, and fabrication outputs were developed in Altium Designer. The custom PA circuit was simulated in LTSpice. The firmware of the device was developed in Code Composer Studio by Texas Instruments and built on top of APIs provided by its embedded SDK for the CC1310 microcontroller.

b. Design Constraints

In addition to the design constraints described in the previous sections, we also received additional design guidelines from Dr. Khalifa:

- The device has to use a thin 3.7 V Li-ion battery and be able to withstand at least several hours of continuous operation without charging.
- The output signal frequency needs to be software-adjustable.
- The output power needs to be software or hardware-adjustable (potentiometer/switch).
- The PCB cannot exceed 70 mm x 50 mm dimensions.

c. Engineering Standards

The main engineering standards we followed pertained to RF electronics. It is an industry standard to match all inputs and outputs of RF circuits to 50 ohm load impedance, and to use special cables called SMA to connect RF signals between PCBs and measuring equipment to maintain matching.

6. Needs and Impact Analysis

The DTCP transmitter-recorder device is designed to serve the researchers at Neural Interface Technology Lab by completing a system they designed. By integrating multiple functions into a single compact PCB, the project reduces the need for several discrete and costly lab instruments, lowering the overall financial burden on research facilities. Furthermore, concerning environmental impact, the system reduces the material footprint associated with manufacturing and maintaining multiple separate pieces of lab equipment. The project's broader societal impact lies in supporting the development of fully injectable, wireless electroceutical devices that can enable minimally invasive neural interfacing for therapeutic applications [15].

Potential risks associated with this device primarily relate to electromagnetic safety, hardware reliability, and ethical compliance. DTCP operates in the MHz range, where tissue heating and unintended stimulation can occur if current density exceeds safe limits [15]. Additionally, electromagnetic interference (EMI) may couple into nearby circuits or distort bioelectric recordings if insufficient shielding or grounding is not implemented. Hardware faults could result in unstable power output, which may damage the implanted device or produce misleading data. Ethical risks include compliance with the Institutional Animal Care and Use Committee (IACUC) standards for all animal procedures [15]. By prioritizing adherence to IACUC standards and non-invasive power delivery, we advance human health while ensuring the humane treatment of research animals.

7. Results

Since the first version of the system was composed of several partially assembled PCBs, we were able to test and develop the subsystems in parallel.

RF Transmitter:

For the alpha build, the first stage and the second stage were on separate boards for isolation and easier debugging. We tested the output power and the output frequency and they were measured using a spectrum analyzer at each stage of the power amplifier. In the process of testing, we determined that we need to increase the supply voltage of the first-stage PA to allow it to produce enough power for the second stage to amplify it to the intended 30 dBm. Initially, we tested 5V - 9V, and determined that at 13V, the first stage produces power of 10.67 dBm, which then is amplified to 30.67 dBm, taking into account the 20 dBm attenuator used to protect the spectrum analyzer.

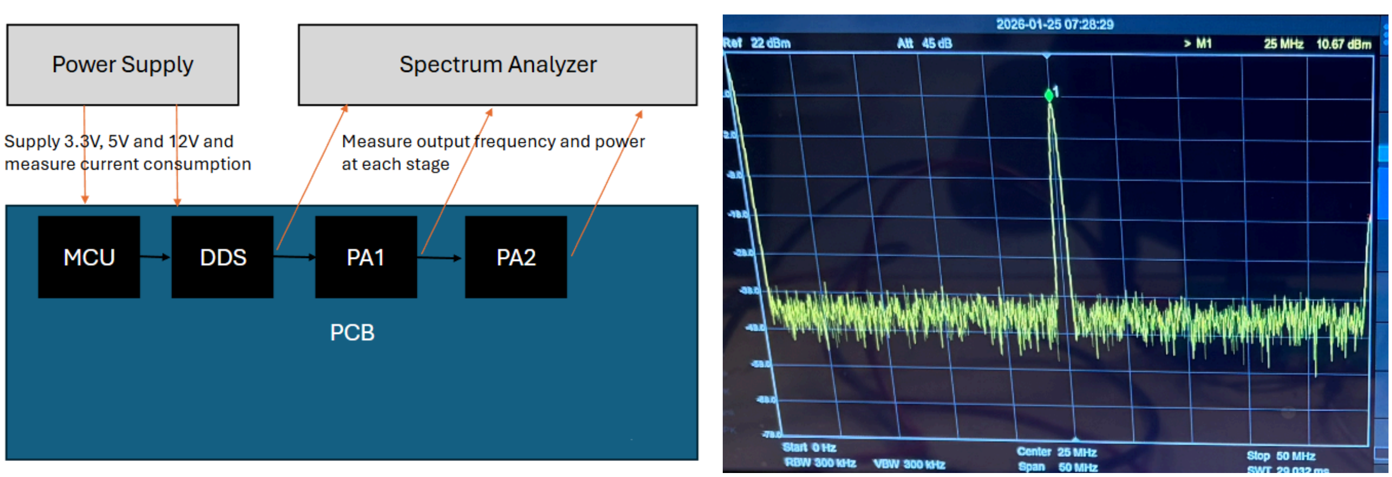
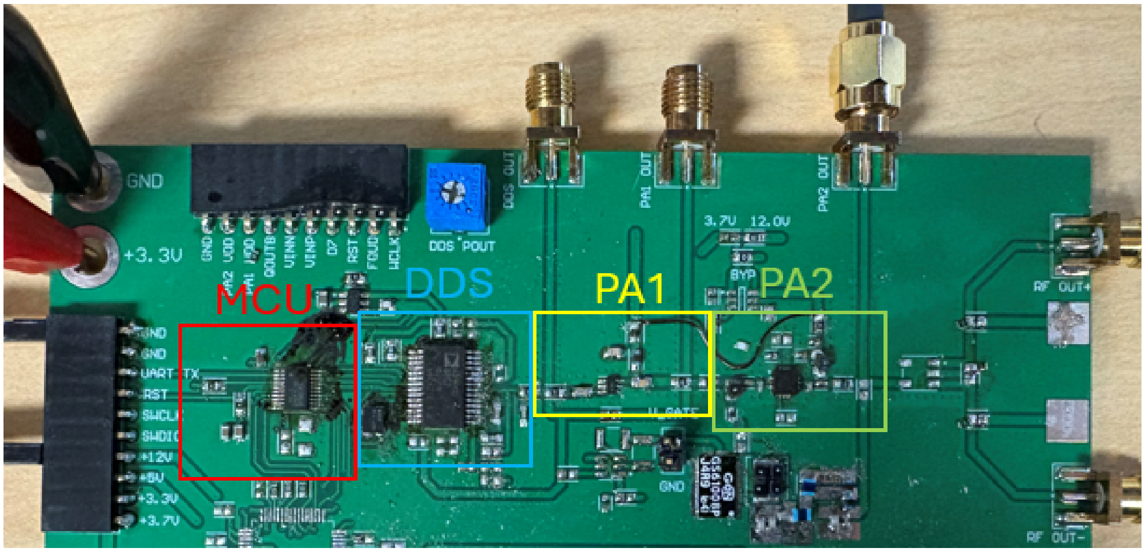


Figure 16: Testing methodology for the RF circuit and the final reading of output power meeting the requirement of 1W (with a 20 dB attenuator in series, bringing the signal from 30.67 dBm down to 10.67 dBm)

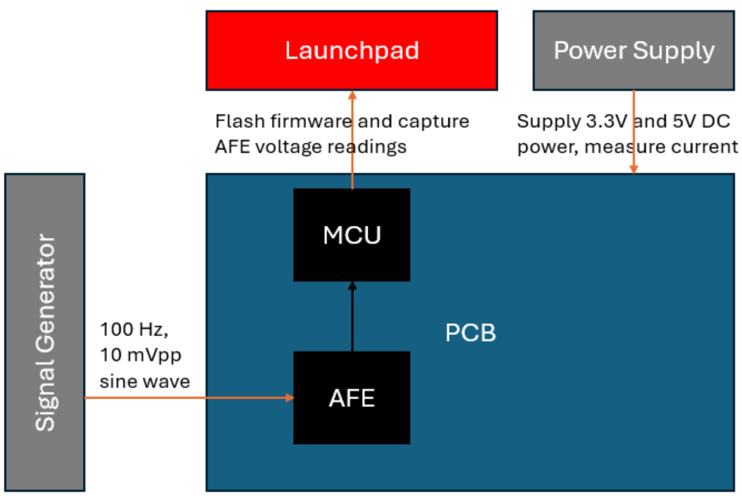
For the beta build, we focused on testing the RF Transmitter, but with all the components on a single board. It followed the same general testing methodology as with the alpha build.



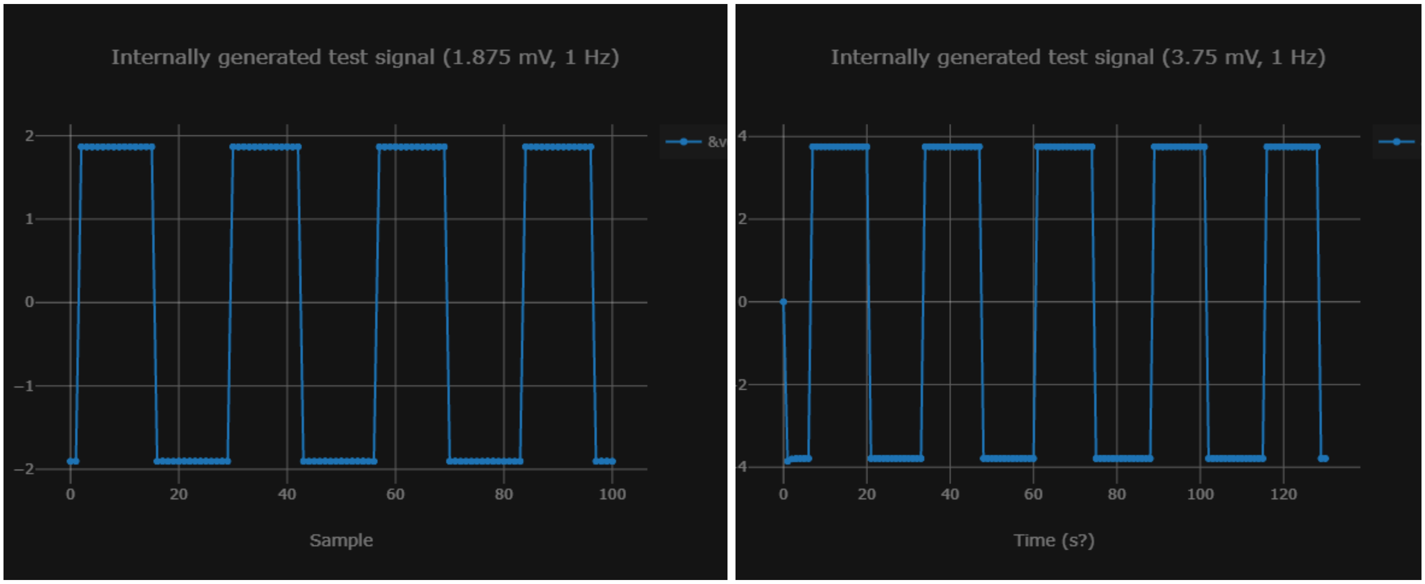
RF transmitter portion of the device (MCU + DDS + PA) integrated on one board

EMG Recording:

For the alpha build, because initially we were not able to properly measure the input signal, even as we changed the amplitude and/or frequency (essentially measuring noise), we wanted to pin down whether the problem was with the chip itself or with something else, like the input channels or the way data or communication (SPI) is handled by the microcontroller. After going through the datasheet, we found an option to internally generate a test signal to see whether the device was working properly, bypassing the external connections, which allowed us to affirm that the device and the SPI communication were working correctly and limit the problem to the external input.

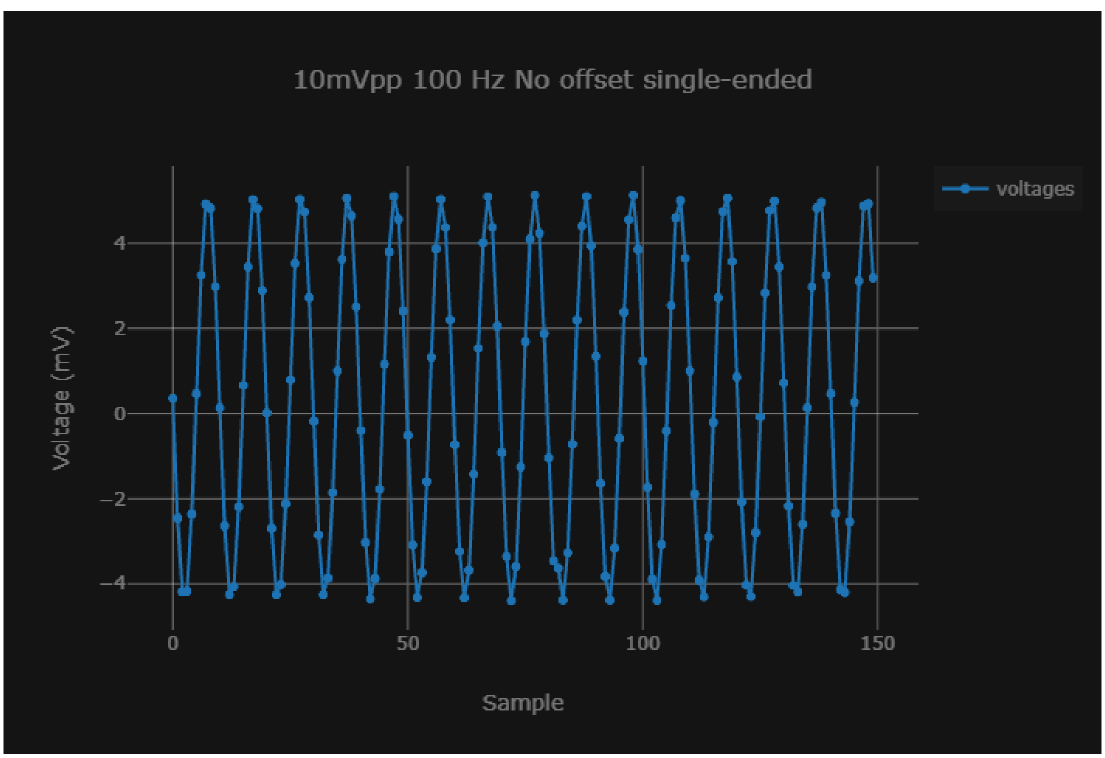


EMG Testing setup for the alpha and beta builds



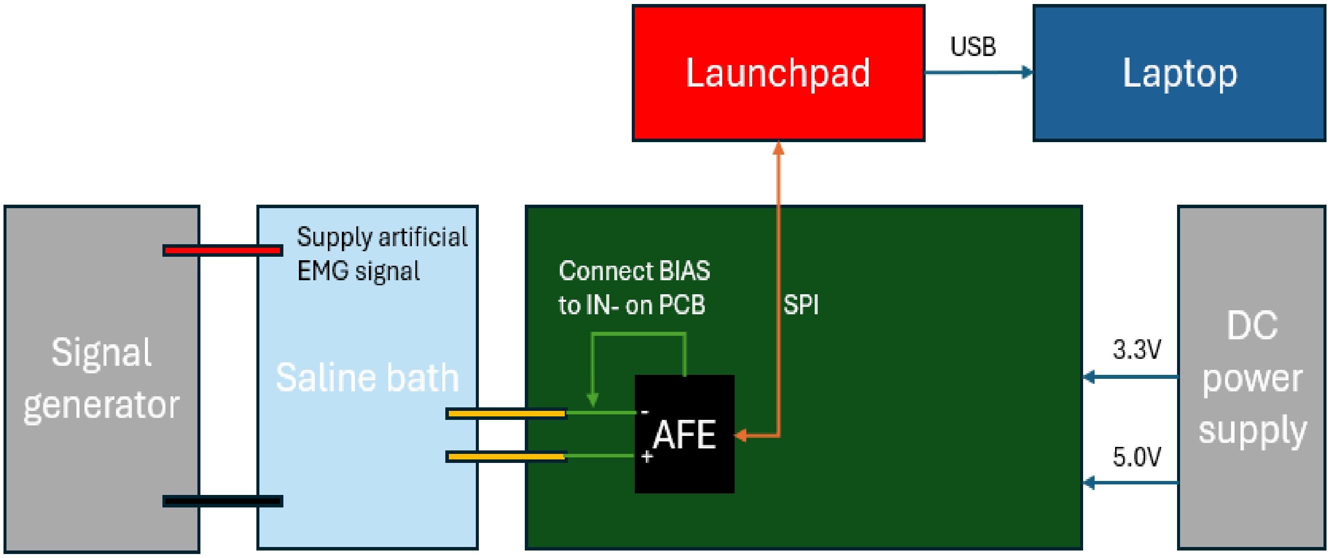
Measurements with internally generated test signals with different amplitudes

With the beta build, we were able to determine that some sort of DC biasing (with bias of mid-supply, which in our case is 2.5V, yielding the best results) is necessary for the device to properly measure signals. After applying a DC offset to the signal, we were able to properly measure the external input signal with the frequency and amplitude of the signal we intend to capture. However, applying a DC offset through the function generator does not meet the working conditions that the device is intended to operate in, so we had to achieve that offset internally. After testing different connections between the channel input and internal bias, we were able to properly capture the input signal, which can be seen in the plot below. We did not account for biasing in the initial version of the design, so we fixed this in the next iteration.



Measurement of an external input signal with no DC offset from the function generator

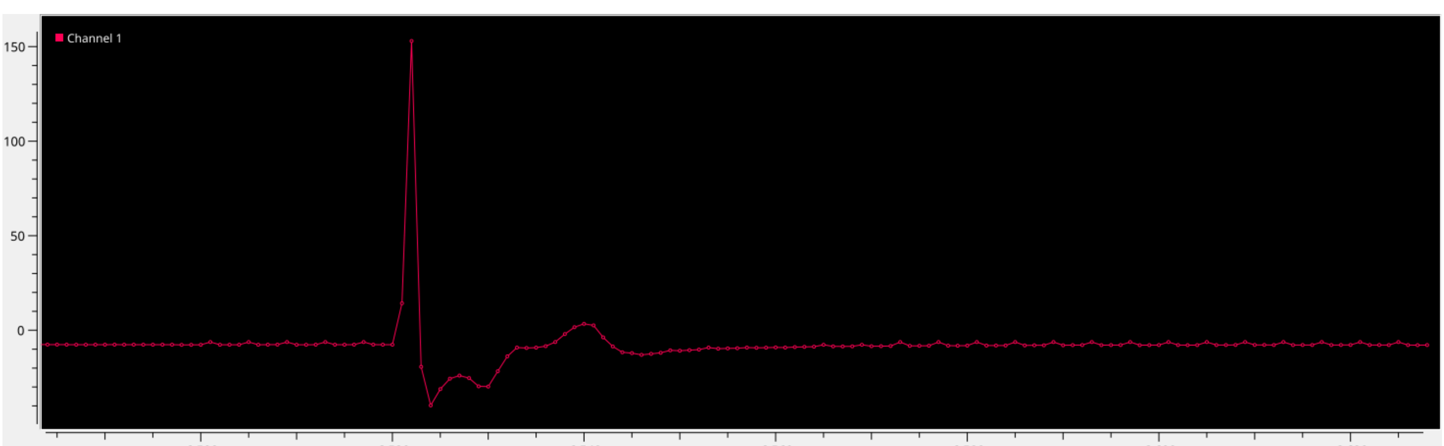
Afterwards, to better mimic the setup in which the device will actually be used, we started to use a saline solution. The way it fits the workflow can be seen below.



Testing setup with saline

We also started to use UART to transfer the data between the microcontroller and the host computer, instead of plotting within the IDE, to get real-time measurements. We encountered some issues that were not present when we were using the Graph functionality of CCS (specifically, spikes that are much higher than the rest of the data). After increasing the baud rate and changing how UART sends data from ASCII to simple binary, the spikes disappeared, and we were able to properly receive the data.

After testing the setup with a test EMG signal generated by custom Python scripts and loaded into the function generator, we determined that our initial AFE sampling rate of 1 kHz was not enough for our purposes. However, when we increased the sampling rate to the maximum of 16 kHz, we again encountered the issue of spiking. We determined it was due to synchronization issues between UART and SPI, with UART not sending data fast enough (e.g., with UART baud rate of 115,200 bps, which was what we had before, and each sample being 32 bits, at best conditions we would be able to handle at most 3600 samples per second). After numerous tests of different configurations of the sampling and baud rates, we settled on the sampling rate of 8 kHz and the UART baud rate of 800 kbps, which seems the fastest we can do with no spiking.

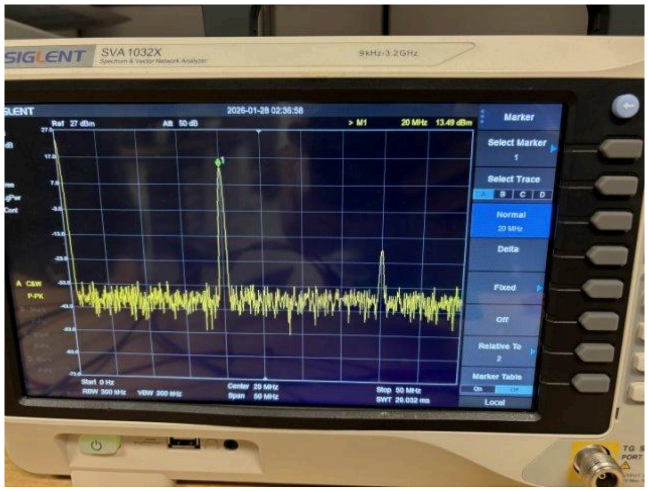


Measurement of a test EMG signal preceded by a stimulation artifact with 8kHz sampling rate in saline.

Class E PA:

One of the designs for the power amplifier that was tested was the Class-E PA. We believed that this could be an ideal solution as Class-E circuits operate a single transistor, and it does so as a switch rather than a current source. A properly designed load network ensures that the transistor turns on only when the drain voltage is near zero (Zero-Voltage Switching, ZVS) and the drain current is minimized at turn-off (Zero-Current Switching, ZCS). Under ideal conditions, this eliminates simultaneous high voltage and high current in the device, reducing switching losses dramatically compared to other PA classes [16].

Upon further testing, we found that our current design fails to meet the demands of the device we are designing.

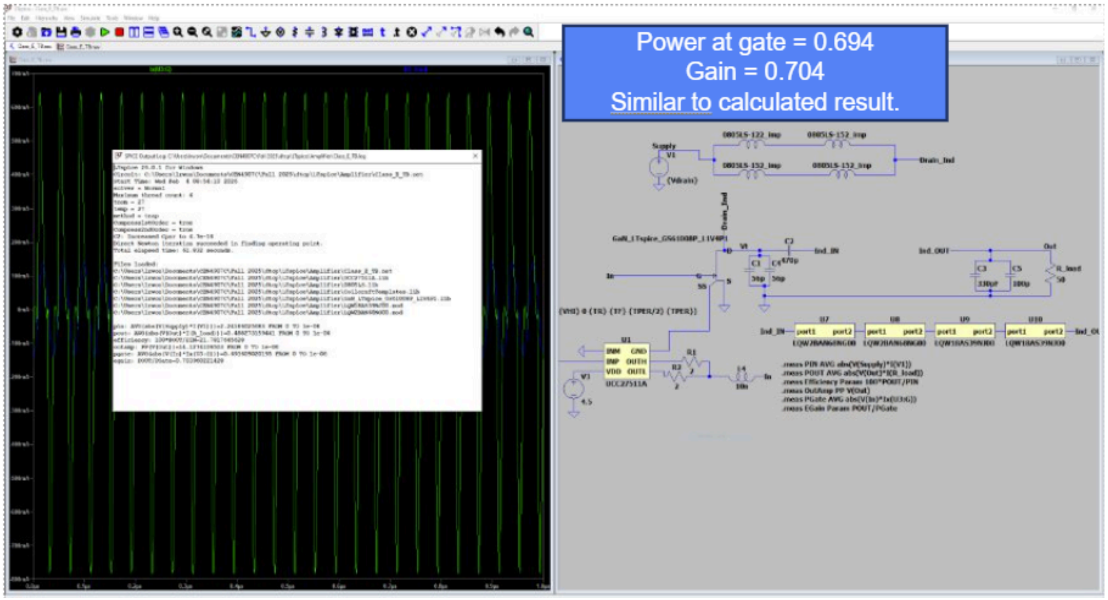


13.5 dBm output power is confirmed here with no attenuator in series. This is our best result in the lab with the current Class E setup falling short of the >= 30dbm we want.

Through testing, we found that generating a 0 to 4.5V square wave produced the best results when input at the gate of the transistor we used. Even at its best, the best power we could output was 13.5 dBm. This was found by sweeping the frequencies with the spectrum analyzer, where we found the device peaks at around 18-20Mhz which is short of the 25MHz we wanted the device to amplify.

Performance metric	Target value	Measured value as of 1/28/2026	Explanation
Output frequency	25 MHz	25 MHz	Measured with an RF spectrum analyzer.
Output power	≥30 dBm differential (1W)	7.26 dBm (0.0053 W)	Measured with an RF spectrum analyzer.
Static current	<10 mA	0 A (without power switches or driver)	Current consumed when the RF signal is not being transmitted. Class E PAs drive the transistor fully on and off like a switch. If the switch is off no DC path from supply to ground.
Tx current	<500 mA	1.37 A	Current consumed when the RF signal is being transmitted.

Further simulation and research show that there was an issue with how we fundamentally designed the Class E PA, in that when choosing our transistor, we failed to take into account the gate charge affecting the gain we would theoretically be able to get. This is also confirmed by LTspice, which compared the power going into the gate of the transistor to the power at the output of the Class E PA. As a result, due to time constraints, we elected to drop this design in favor of the Class A PA design, which met the target values given above.



Screenshot of LTspice simulation showing the class E PA attenuating when using real parts (recreation of lab setup), resulting in a gain of less than one.

8. Status

For the RF transmitter:

Feature	Completion Status	Details	Timeline
Class-A PA	Done	The circuit is working correctly, achieving the specified specs.	January 2026
Class-E PA	Abandoned	Class-E PA was a custom circuit that may have been more efficient compared to Class-A, but we did not get the expected results and decided to discard the design and focus our efforts on the Class-A PA	February 2026
Power Switching	Done	Can power switch DDS and power amplifiers to reduce idle current consumption.	March 2026

For the EMG recording sub-system:

Feature	Completion Status	Details	Timeline
SPI Communication between the AFE and the MCU	Done	The microcontroller can write to and read from the AFE with no synchronization issues. The AFE responds to SPI commands, returning correct values in accordance with the datasheet, and sends measurement data in the correct order.	January 2026
EMG Measurements with an SNR of 10 dB or more	Done	We determined that biasing is necessary, and after connecting the bias and the input signal (whether through saline or through a negative input channel terminal), we can properly measure the input signal, meeting the SNR specs.	February 2026
Real-time Measurements with UART	Done	The system is working, and we can receive data (samples from AFE) from the MCU and plot in real-time, though there are still some things related to synchronization that could be improved (e.g., add a starting header byte, though that would incur additional overhead).	March 2026
Persistent State	Done	The software we are using to plot the measurement data (SerialPlot) sent from the AFE to the PC gives an option to save that data in a file.	March 2026

9. Conclusion

The final system architecture will deliver a 1W pulsed signal for neural stimulation, maintain a low standby current of 1mA, and collect data to plot in real-time, all integrated in a compact 65mm x 48mm PCB. By using the verification methods stated above, using a saline medium, we were able to confirm that the device meets the primary objectives of powering thread-like implants and monitoring real-time physiological responses.

As we move into the final phase of our project, we will focus on receiving the new PCB version and consolidating the firmware into a single, unified codebase. Upon final validation through animal experiments, a copy of the device will be delivered to Harvard University, providing them with a cost-effective and streamlined platform for advancing in vivo neural interface research.

10. References

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